Flame Freefall

Your AI research companion
for NASA combustion evidence.

Ask in natural language. Inspect cited evidence.
Compare experiments. Export your research dossier.

Code Nashai / Muhammad Marjan Ahmed / Solo builder

Prevention

Material flammability

Detection

Smoke and sensors

Suppression

Droplet extinction

The hard part is knowing what transfers.

A spacecraft-fire researcher asks:

Which experiments inform a study near a proposed Lunar atmosphere?

Find

Tables and reports are spread across investigations.

Compare

Matching oxygen alone does not match the experiment.

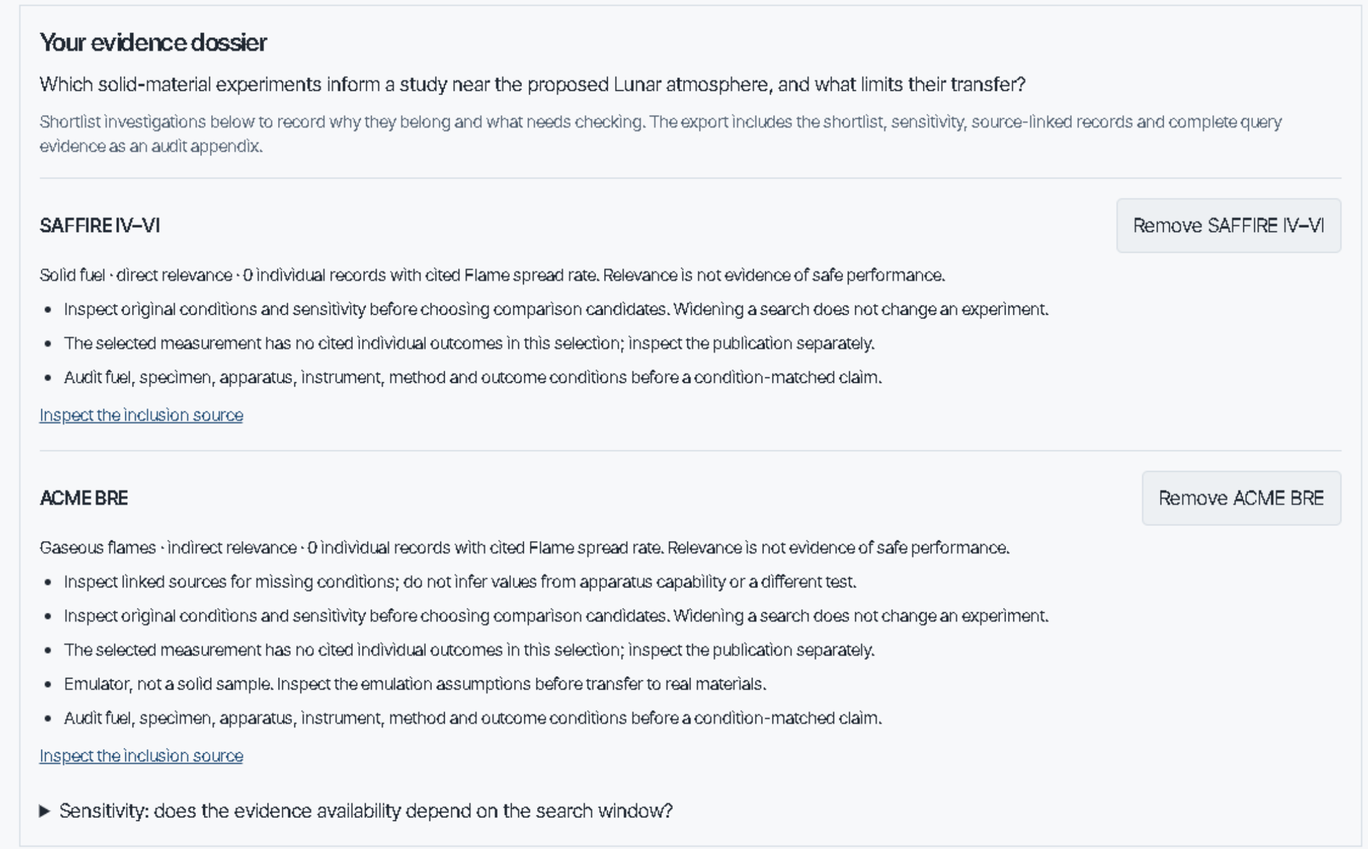
Interpret

A nearby result can still have limited applicability.

The opportunity: ask an AI companion, then inspect the evidence behind the answer.

Source: supplied challenge-details.md; NASA PSI investigations and governed source reports

Ask in chat. Investigate in the workbench.



Actual local dossier capture, cropped from the saved workflow run; not a live deployment check.

1 Ask the AI chatbot

Search research and test records.

2 Inspect the evidence

Check conditions and source limits.

3 Build your dossier

Use the workbench to shortlist and export.

Implemented: ChatKit UI and PSI tools; /decide dossier; query and comparison twins; docs/evidence-method.md

From a question to evidence you can inspect.

Ask the AI chatbot

Show solid material tests near a Lunar atmosphere.

SAFFIRE IV–VI

Direct solid-fuel evidence.
Inspect setup and missing fields.

ACME BRE

Indirect condensed-fuel emulation.
Keep its physical geometry distinct.

Widening the oxygen search window changes the retrieved evidence.

The experiment has not changed. Neither has its transfer limit.

Compare with evidence. Keep the context.

77°C

VI-3 / two-sided PMMA

57°C

VI-4 / one-sided PMMA

Reported observations from Table 2

Chat + Explorer audits

Fuel description differs.
Oxygen and pressure differ.
Method and dimensions remain incomplete.

Inspect both results. Do not infer a controlled effect.

Meet your AI combustion research companion.

Find and understand

Ask questions in natural language.

Retrieve cited research documents.

Find tests by fuel and conditions.

Open individual records and sources.

Explore and compare

Plot measured outcomes.

Inspect gap maps and atmosphere coverage.

Compare investigations and test records.

Order eligible individual measurements.

Natural-language question → research tools → source-linked widgets

Code renders PSI measurements; narrative research answers carry source citations.

1,951 indexed records / 1,838 test or test-point records / 23 stable documents

Missing metadata stays visible. Coverage and measurement ordering do not establish safety.

Ask better questions. Keep the evidence.

For researchers and aerospace engineers

An AI research conversation.
A source-backed comparison.
A reproducible evidence dossier.

Next: prove the benefit

Observe researcher task completion.
Review relevance with domain experts.
Resolve critical source metadata.

Explore Flame Freefall

AI chatbot + Explorer + evidence dossier

From NASA combustion data to questions worth investigating.

Designed to support research; measured time savings and safety outcomes are not established.